

Technology Stack

Project: Automated Network Request Management (ServiceNow)

1. Platform

Component	Details
Platform	ServiceNow Personal Developer Instance (PDI)
Application Scope	Custom scoped application: Automated Network Request Management
Hosting	Cloud-hosted (developer.servicenow.com)

2. ServiceNow Modules Used

Module	Purpose
Service Catalog	Request intake form with variables (Request Type — VPN Access / Firewall Rule / New Connectivity / Guest Access, Justification, Device/System, Network Segment, Required By Date)
Flow Designer	Approval routing, branching logic (Approved/Rejected), task creation, and email notifications
Business Rules	Server-side automation — e.g., auto-creating an Incident on capacity or security policy conflict
Approvals Engine	Native ServiceNow approval tracking (sysapproval_approver) and My Approvals interface
Access Control Lists (ACL)	Role-based read/write restrictions on request records
Studio	Scoped application development environment used to build and manage all components

3. Scripting & Logic

- Server-side scripting: JavaScript (GlideRecord, GlideDateTime APIs) within Business Rules
- Flow Designer: low-code visual workflow logic with data pills for dynamic field references

4. Data Storage

Table	Purpose
sc_cat_item	Catalog item definition and variables
sc_request	Parent request record
sc_req_item	Individual requested item — core record for approval and fulfillment tracking
sysapproval_approver	Approval records assigned to managers
sc_task	Fulfillment task assigned to the Network/IT Support team
incident	Auto-created when a request cannot be fulfilled (e.g., capacity or security constraint)

5. Version Control & Documentation

- Source Control: GitHub (Update Set XML export / Studio source control integration)
- Documentation: Markdown (README.md) and Microsoft Word / PDF for supporting project documents

6. Tools Used During Development

- ServiceNow Studio — application and component development
- Flow Designer — visual workflow builder
- Postman / Browser DevTools — optional, for debugging (if used)
- Git & GitHub — version control and project publishing